RHM sampling complexity: a third depth, in 1D

exp(Theta(n)) holds. P* = 0.21|R_L| is a 2D statement, not a universal one.

113 grammars x 19 values of P at the root-reaching window k = d - 1,
depths n = 4, 8, 16. 110 peaks bracketed.

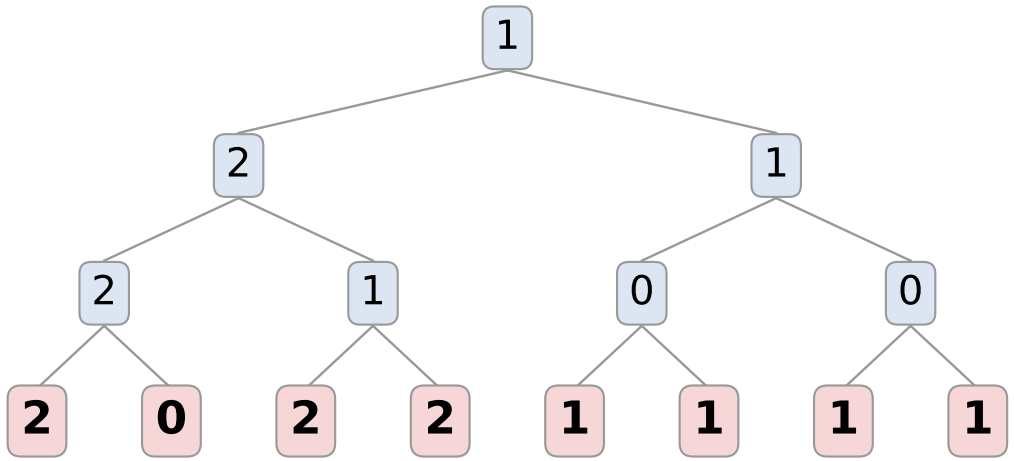
ln P* grows 0.374 per site: P* = exp(Theta(n)), now at a depth
2D cannot reach (L=4 would need a 15x15 patch).

But P* ~ |R_L|^0.689 +- 0.045, 6.8 sd below the 2D exponent of 1.

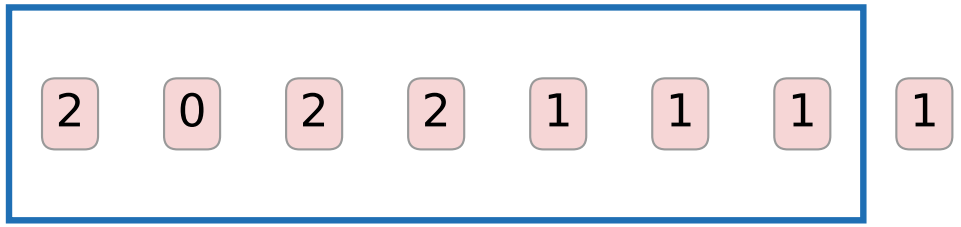
Matched-n control: at n = 16, x* = 0.050 in 1D vs 0.193 in 2D,
while 2D is flat in n. The difference is dimension, not size.

1. Tokenization: the string, the tree, the window

v=3, s=2, L=3: root -> 8 leaf tokens



the sample the machine actually sees

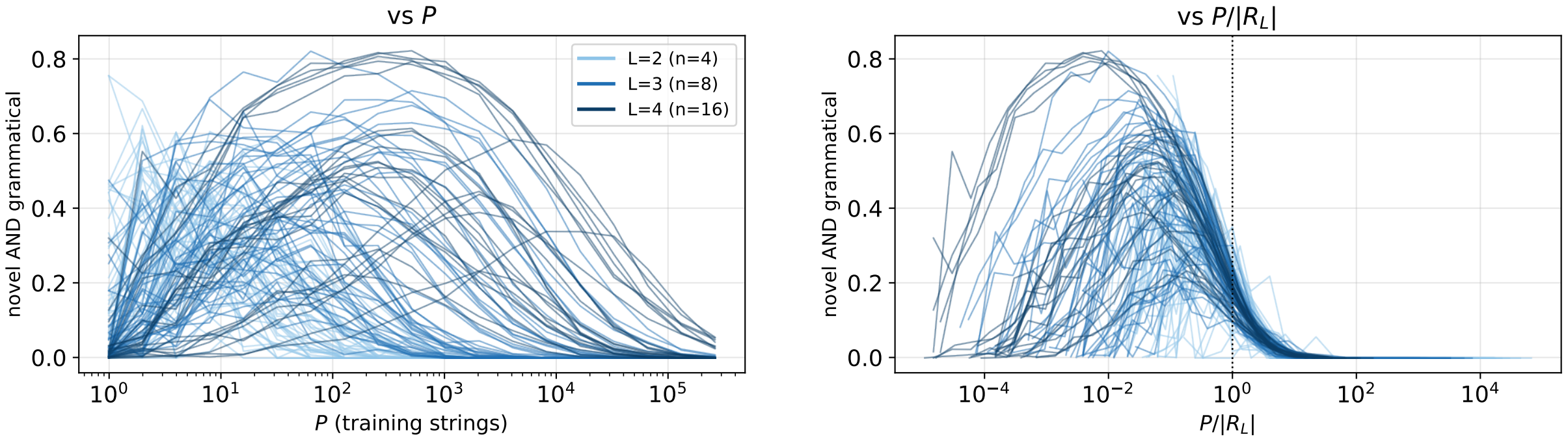


one U=1 window, k = 2s^(L-1) - 1 = 7

covers 7 of 8 sites = 88% of the sample

A token is one leaf symbol from the alphabet 0..v-1. One *sample* is the whole length-d string, d = s^L. The machine is a lookup table over length-k windows: for a masked site it returns the empirical distribution of the centre tokens of every training window that agrees on all observed sites. Off-grid sites get their own marker symbol v and count as observed, so a window at the border can only match training windows framed the same way.
k = 2s^(L-1) - 1 is the smallest window that fully contains the top-level block, i.e. the smallest one that can see the root. In 1D that is d - 1 at every depth, which is the whole story of this deck.

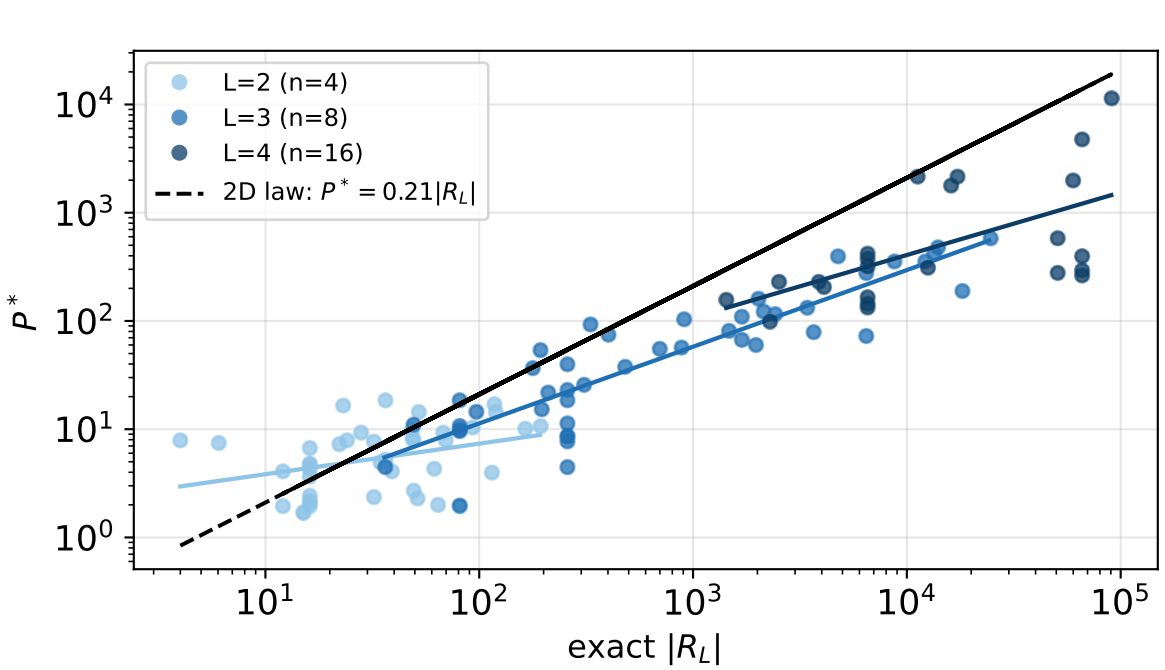
2. Success peaks, and what collapses them

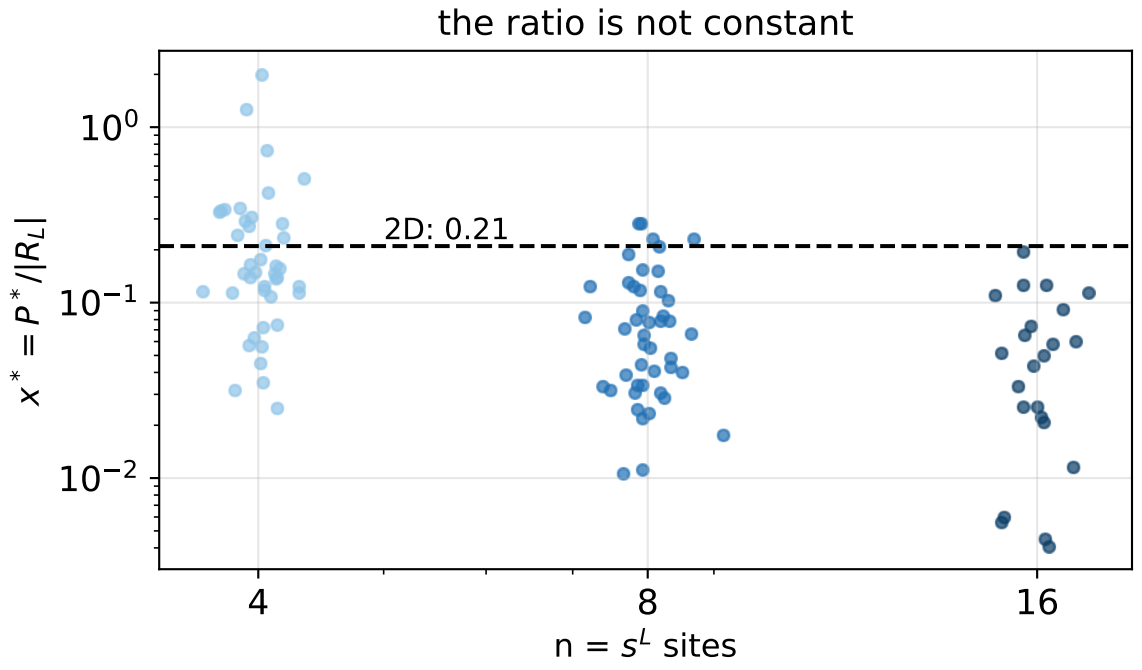


Each curve is one grammar (v, m, rule seed) swept over a 2x ladder in P from 1 to 262144. Success is hshunt's criterion: the generated string is not in the training set AND parses all the way to a root symbol. The peak is real -- grammaticality needs the table filled, novelty needs the training set to be a small part of the grammar, and they pull opposite ways.

In 2D, dividing by the exactly-enumerated |R_L| collapsed these curves onto one. Here it visibly does not: the three depths stay separated, and that separation is the result.

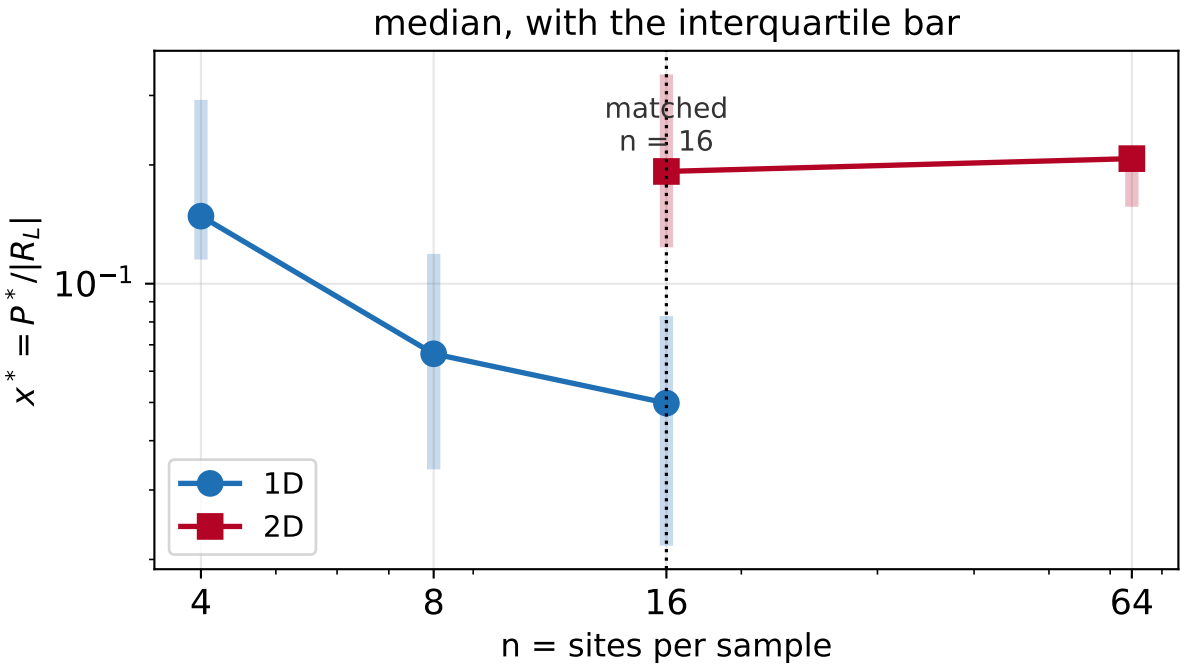
3. $P^* \sim |R_L|^b$ with b well below 1





Left: each point is one grammar's bracketed peak against its exactly enumerated |R_L|. If P* were Theta(|R_L|) the points would lie on the dashed line. Free fits per depth: L=2: b = 0.283, L=3: b = 0.708, L=4: b = 0.579; pooled b = 0.708 +- 0.028. Dropping cells at the ladder floor (P* < 8, where an integer P on a 2x ladder cannot resolve a slope) and the n=4 depth leaves 64 cells and b = 0.689 +- 0.045 -- 6.8 sd below 1.
Right: the same data as the ratio. In 2D this scatter was flat at 0.197 with a 1.95x geometric spread. Here it falls with depth.

4. Matched-*n* control: it is dimension, not *n*



at n = 16, both dimensions:

1D (L=4, k=15): $x^* = 0.050$

2D (L=2, k=3): $x^* = 0.193$

3.9x apart

and 2D is flat in n:

0.193 at n=16 vs 0.207 at n=64 (1.08x over 4x in n)

This is the control that makes the result interpretable. 1D at L=4 and 2D at L=2 have the same number of sites, so if x* were a function of n they would agree. They differ by 3.9x, while 2D itself barely moves across a 4x change in n. The fall is therefore not an n effect, and a joint power fit across both dimensions (which would have returned x* ~ n^-0.154) is meaningless -- it averages a flat series against a falling one. That fit is retracted.

What does differ at matched n is coverage: one U=1 window spans 15/16 = 94% of a 1D string but only 9/16 = 56% of a 2D grid. The 1D machine is far closer to whole-string lookup. Consistent with that, its median peak success is 0.49 against 2D's 0.022. This is a hypothesis the deck does not yet test -- see the last page.

5. Three real traces: reveal-one denoising, v2 m2 L=3

P = 8 training strings, |R_L| = 49 exactly, k = 7, alpha = 1e-3. "." = still masked.

step	state	site	posterior	->
0	....	4	[0.50 0.50]	1
1	....1...	3	[0.67 0.33]	0
2	...01...	5	[0.50 0.50]	1
3	...011..	7	[1.00 0.00]	0
4	...011.0	1	[0.50 0.50]	1
5	.1.011.0	6	[0.50 0.50]	1
6	.1.01110	0	[0.50 0.50]	0
7	01.01110	2	[0.50 0.50]	1

final 01101110
grammatical: True
novel: True
=> SUCCESS

step	state	site	posterior	->
0	....	1	[0.00 1.00]	1
1	.1......	2	[0.25 0.75]	1
2	.11.....	4	[0.86 0.14]	1
3	.11.1...	6	[0.00 1.00]	1
4	.11.1.1.	7	[1.00 0.00]	0
5	.11.1.10	0	[0.83 0.17]	0
6	011.1.10	5	[1.00 0.00]	0
7	011.1010	3	[0.00 1.00]	1

final 01111010
grammatical: True
novel: False
=> no

step	state	site	posterior	->
0	....	0	[0.87 0.13]	0
1	0.......	7	[0.50 0.50]	1
2	0......1	5	[0.00 1.00]	1
3	0....1.1	1	[0.00 1.00]	1
4	01...1.1	4	[1.00 0.00]	0
5	01..01.1	2	[0.20 0.80]	0
6	010.01.1	6	[0.50 0.50]	1
7	010.0111	3	[0.00 1.00]	1

final 01010111
grammatical: True
novel: False
=> no

One uniformly random masked site per step; the posterior is the empirical centre distribution over the training windows that agree on every observed site, Laplace-smoothed. Early steps see almost nothing observed, so the posterior is close to the marginal; late steps are nearly determined. A string is scored only at the end, and needs both grammaticality and novelty to count -- which is why P sits at 8 here and not higher: at large P the machine reproduces its training set.

6. Every depth, every grammar family

L	n	k	v	cells	|R_L| range	P* range	med x*	succ
2	4	3	2	12	4-16	2-8	0.196	0.548
2	4	3	3	15	12-64	2-18	0.166	0.428
2	4	3	4	15	22-192	2-17	0.118	0.446
3	8	7	2	15	36-256	2-23	0.091	0.509
3	8	7	3	15	176-6400	15-123	0.066	0.451
3	8	7	4	15	208-24298	22-576	0.043	0.499
4	16	15	2	15	2304-65536	102-592	0.026	0.565
4	16	15	3	5	1429-65480	158-4821	0.092	0.389
4	16	15	4	3	11130-90368	2182-11499	0.127	0.338

x* = P*/|R_L| with P* from a log-P parabola through the top three ladder points, reported only when the argmax is interior; 3 of 113 cells were at the ladder edge and are excluded rather than clamped. succ is the peak success itself. |R_L| is enumerated exactly, never taken from the nominal v m^M, because the rules are drawn with replacement and the collisions are what give the collapse test its range.

7. Against the previous best on the same eval

	2D sweep (prev, 11-03)	1D depth probe (this run)
bracketed peaks	30	110
depths	L=2,3 (n=16, 64)	L=2,3,4 (n=4, 8, 16)
window	k=3, 7 (2D patch)	k=3, 7, 15 (1D run)
median x* = P*/|R_L|	0.197	0.083
free fit b	0.965 +- 0.041	0.689 +- 0.045 (robust)
b vs 1	0.9 sd -> consistent	6.8 sd -> ruled out
median peak success	0.022	0.490

WHAT SURVIVES. P* = exp(Theta(n)) is confirmed at a third depth: ln P* grows 0.374 per site over n = 4..16, and since |R_L| = Theta(v m^((n-1)/(s-1))) is itself exponential in n, any positive power of it is too. The deadline deliverable stands.

WHAT DOES NOT. The sharper 2D statement P* = 0.21|R_L| is 2D-specific. In 1D the exponent is 0.689 +- 0.045 and the constant is 2-4x smaller, and the matched-n control on page 4 shows this is not an n effect.

NEXT, AND NOT YET DONE. The coverage hypothesis -- that x* falls as one window covers more of the sample -- is consistent with everything here but untested. The clean test is inside 2D alone: sweep k below the U=1 value and see whether x* tracks k^2/n. If it does, 1D and 2D are one law in coverage, not two in dimension.